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| 标题 | Regioselective Transfer Hydrogenation of Substituted Oxiranes with Alcohols Using MgO as the Catalyst |
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## 摘要（原文节选）

An excellent regioselectivity of vapor-phase catalytic transfer hydrogenation (CTH) of substituted oxiranes (methyl-, n-butyl-, and phenyloxirane) with alkanols (EtOH, 2-PrOH or 2-PeOH) as hydrogen donors in the presence of magnesium oxide as a catalyst was attained. In the vapor phase, all of these oxiranes as well as 1,2-epoxycyclohexane were hydrogenated. Moreover, it was found that primary alcohols were always the main products of CTH of methyl-, n-butyl-, and phenyloxirane with very good regioselectivity towar ⋯

## 数据表 · Table 1（17 行，含数值行 16 行）

Vapor-phase transfer hydrogenation of n-butyloxirane with 2-propanol in the presence of magnesium oxide as the catalyst. Influence of donor – acceptor molar ratio on the composition of the post-reaction mixture.

| D/A | T[K] | Conv.[%] | Yields of Products 1 [%] | | | |
|---|---|---|---|---|---|---|
| Me-CO-n-Bu | 2-HeOH | 1-HeOH | Others | | | |
| 3 | 523 | 1 | 0 | tr | 1 | tr |
| 573 | 4 | 0 | 1 | 2 | 1 | |
| 623 | 38 | 12 | 6 | 13 | 7 | |
| 673 | 84 | 12 | 18 | 43 | 11 | |
| 6 | 523 | 2 | 0 | 1 | 1 | tr |
| 573 | 7 | 0 | 3 | 3 | 1 | |
| 623 | 49 | 10 | 12 | 21 | 6 | |
| 673 | 84 | 15 | 19 | 42 | 8 | |
| 9 | 523 | 2 | 0 | 1 | 1 | tr |
| 573 | 7 | 0 | 3 | 4 | tr | |
| 623 | 39 | 10 | 10 | 17 | 2 | |
| 673 | 91 | 14 | 20 | 54 | 3 | |
| 6 2 | 523 | 2 | 0 | tr | 1 | 1 |
| 573 | 6 | 0 | 1 | 3 | 2 | |
| 623 | 23 | 12 | 2 | 5 | 4 | |
| 673 | 45 | 17 | 6 | 13 | 9 | |

脚注（逐字）：1—Me-CO-n-Bu = 2-hexanone; 2-HeOH = 2-hexanol; 1-HeOH = 1-hexanol; 2—ethanol as hydrogen donor.

正文对应句（逐字）：It was found that for the 2-pentanol – methyloxirane mixture with a molar ratio of 3 (Figure 1a), the conversion increased with temperature reaching the highest value of 41%. In the entire studied reaction temperature range (473 – 623 K), oxirane reduction took place, and the primary alcohol, i.e., 1-pr

## 数据表 · Table 2（8 行，含数值行 7 行）

Time-on-stream test of vapor-phase transfer hydrogenation of n-butyloxirane with 2-propanol in the presence of magnesium oxide as the catalyst. D/A = 6, TR = 673 K.

| Time[h] | Conv.[%] | Yields of Products 1 [%] | | | |
|---|---|---|---|---|---|
| Me-CO-n-Bu | 2-HeOH | 1-HeOH | Others | | |
| 0.5 | 87 | 15 | 22 | 40 | 10 |
| 1.0 | 80 | 12 | 19 | 43 | 6 |
| 2.0 | 75 | 11 | 18 | 41 | 5 |
| 3.0 | 74 | 11 | 16 | 44 | 3 |
| 6.0 | 68 | 7 | 13 | 44 | 4 |
| 3.0 2 | 62 | 7 | 11 | 41 | 3 |
| 6.0 2 | 58 | 6 | 10 | 39 | 3 |

脚注（逐字）：1—Me-CO-n-Bu = 2-hexanone; 2-HeOH = 2-hexanol; 1-HeOH = 1-hexanol; 2—repeated test with the spent catalyst.

正文对应句（逐字）：It was found that for the 2-pentanol – methyloxirane mixture with a molar ratio of 3 (Figure 1a), the conversion increased with temperature reaching the highest value of 41%. In the entire studied reaction temperature range (473 – 623 K), oxirane reduction took place, and the primary alcohol, i.e., 1-pr

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